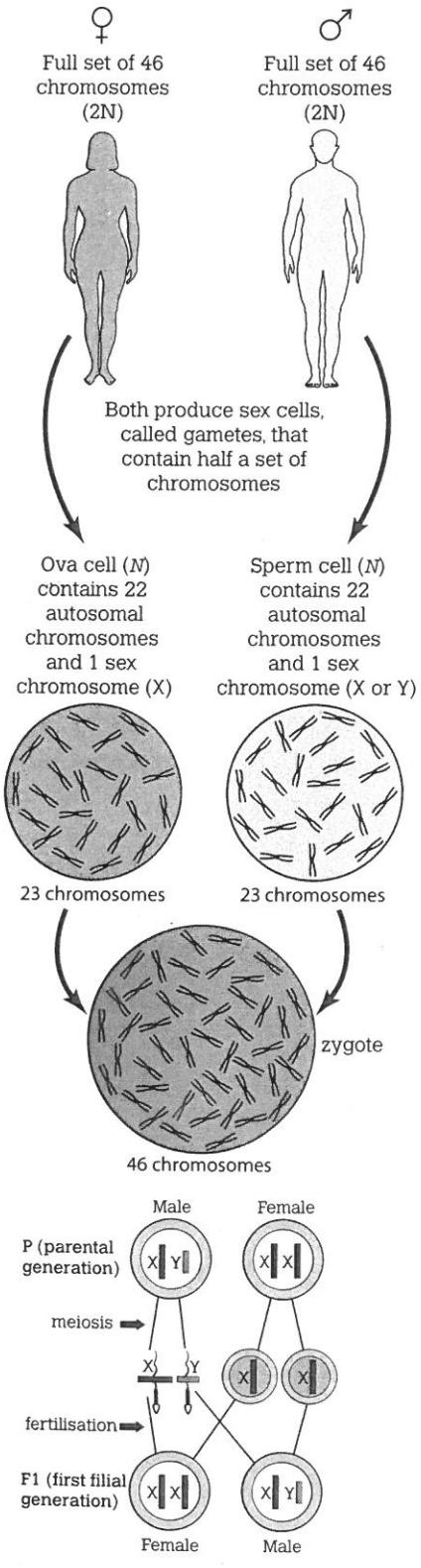
3.3 Understanding genetic basics

Genetics is sometimes called the lottery of life. The genes that are passed from parents to offspring are a random selection of half of the genetic instructions that make up each parent. Together, the two sets of genes provide all of the information that an organism needs to grow and live.



Inheriting genetic information

An organism's **genes** determine its characteristics. It is the organism's characteristics that are then used to classify it into a kingdom, such as Plants or Animals, and into a specific family and species. Genes are also responsible for characteristics that determine how an organism obtains food, its resistance to infection and its behaviour.

Genetic information is passed from one generation to the next in the form of chromosomes. You receive half your chromosomes from your mother and half from your father.

When the two halves combine, they make a full set. A full set is found in every body (**somatic**) cell. This is written as 2*N* (diploid). In the full set in somatic cells, every chromosome has a matching chromosome with the same size, shape, binding pattern and assortment of genes. Together, they are said to be a homologous pair.

In humans, a full set is 23 pairs, or 46 individual chromosomes. Of these 23 pairs, there are 22 pairs of autosomal (body) chromosomes and two sex chromosomes; XX in females and XY in males. **Gametes** (sex cells—the ova and sperm) only contain one chromosome from each of the pairs. This is half a set. Sperm and eggs are said to be haploid (*N*).

Which chromosome is inherited from each pair is completely random. This means different offspring of the same parents will each get a different combination. This is why children in the same family look a little like each other and a little like each parent, but are not identical to them.

Checkpoint 1

Copy and complete the table below with the missing number of chromosomes for each organism. The first has been completed for you.

Organism	Chromosomes in each somatic cell (2*N*)	Chromosomes in each gamete (*N*)
Human	46	23
Apple	34	
Chicken		9
Dog		39
Mosquito	6	

Gender determination

Females have two X sex chromosomes, which means they only produce ova containing X chromosomes. Males have one X and one Y sex chromosome and so a sperm can contain either an X or a Y chromosome. Cell division that produces gametes (meiosis) will put one sex chromosome in each sperm or ovum. It is therefore the male gamete that determines the sex of the offspring.

X (ovum) + X (sperm) = XX female

X (ovum) + Y (sperm) = XY male

The illustration can also be shown in a **Punnett square**. Punnett squares are covered in more detail in Skill 3.4 on page 64.

♂ \ ♀	X	X
X	XX	XX
Y	XY	XY

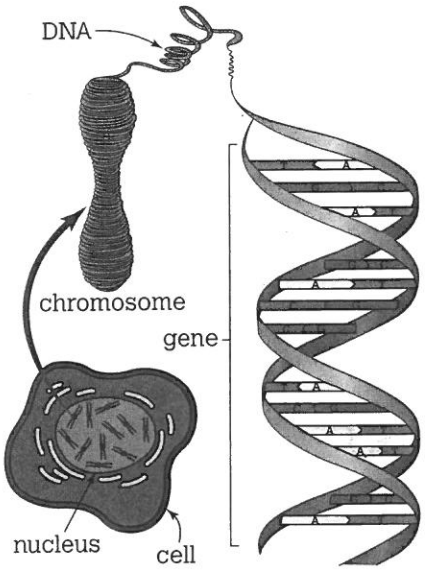
Offspring: probability is that 50% will be XY (male) and 50% will be XX (female).

Checkpoint 2

Propose a reason why the probability (chance) of producing a male is the same as producing a female.

Genes and alleles

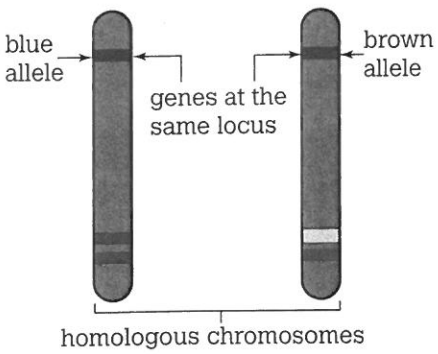
The chemical that stores the genetic code is called **DNA** (deoxyribonucleic acid). Its structure resembles a double helix. Each chromosome contains one very long and tightly coiled strand of DNA. Sections of DNA that contain complete instructions are known as genes. Genes are like 'carriages' along the DNA 'train'.



All human cells (except red blood cells and gametes) contain 46 chromosomes, which include approximately 42 000 genes. Each gene has a specific role in providing information and is only 'turned on' in the cells in which the gene is relevant. For example, the gene for eye colour will only be turned on in the iris of the eye, and not in skin cells.

A homologous pair of chromosomes carries the same genes, in the same places (loci), on each of the chromosomes. There are different variations of every gene that have arisen from mutations. The different variations are called **alleles**. These alleles may be the same on each pair of chromosomes, or different. For example, blue, brown, green and grey are all alleles of the gene for eye colour in humans.

Each somatic cell has two corresponding alleles that code for the same trait. Sometimes, one allele is expressed (revealed) while the other allele is masked (hidden). The allele that is expressed is the **dominant** allele. The allele that is masked is the **recessive** allele.

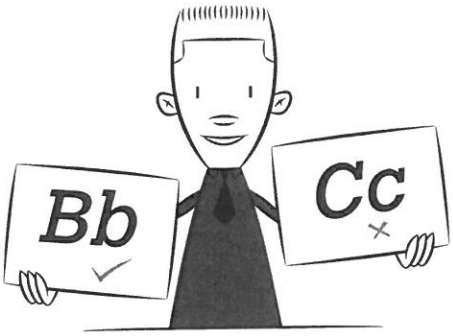


Genotype is the term used to describe the particular combination of alleles that the individual has.

Phenotype is the term used to describe the physical characteristics of an individual that result from the influence of the alleles that are expressed and also the effect of the environment.

A **carrier** is an individual who has one dominant and one recessive allele for a characteristic in their genotype. They carry the recessive allele but it is not evident in their phenotype. The allele can still be passed (carried) on to the next generation.

Alleles are represented by symbols, usually a letter of the alphabet that relates to the characteristic being investigated. For instance, the gene for flower colour may have an allele for red flowers. This is given the uppercase symbol *R* because it is a dominant allele. The recessive allele for white flowers is given the lowercase symbol *r*.

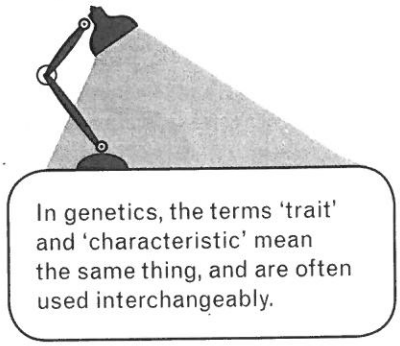


Write letters to represent alleles clearly. Avoid letters that may be confused between uppercase and lowercase, such as C and c.

Checkpoint 3

Copy the following table into your notebook, then **state** a letter symbol suitable to represent each of the alleles for each gene.

Gene	Dominant allele	Symbol	Recessive allele	Symbol
Hairline	Widow's peak		No widow's peak	
Dimples	Dimples		No dimples	
Earlobes	Unattached earlobes		Attached earlobes	
Lips	Broad lips		Thin lips	
Thumb shape	Straight thumb		Hitchhiker's thumb	



In genetics, the terms 'trait' and 'characteristic' mean the same thing, and are often used interchangeably.

Representing different types of inheritance

An organism receives two alleles for each trait. These alleles may be the same or different. The combination of alleles is the individual's genotype.

Individuals with identical alleles at a particular locus of a chromosome are **homozygous** (homo = same). That is, they have two copies of the same allele for the trait.

Individuals with two different alleles for a particular gene are **heterozygous** (hetero = different). They have two different alleles for the trait, and their phenotype for that trait depends on how the two different alleles interact.

The combination of alleles (genotype) affects the organism's phenotype in one of the following three ways.

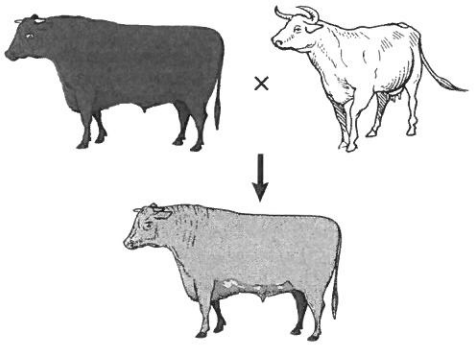
Dominant and recessive alleles

A dominant allele is shown with an uppercase letter, such as *B*. An individual that is homozygous dominant for that gene has two copies of the dominant *B* allele, and so has the genotype *BB*. In this case, both alleles are the same and both are dominant, so 'B' is the trait that is expressed.

An individual that is homozygous recessive for that gene has two identical copies of the recessive allele. This genotype is shown by two lowercase letters, such as *bb*. In this case, both alleles are again the same (because the individual is homozygous) but both are recessive, so 'b' is the trait that is expressed.

A heterozygous individual has two different copies of the gene—one copy of the dominant allele, and one copy of the recessive allele. This is shown by one uppercase and one lowercase letter—*Bb*. In this case, the

dominant allele *B* is expressed in the phenotype and masks the effect of the recessive allele. The individual is a carrier of the recessive *b* allele.

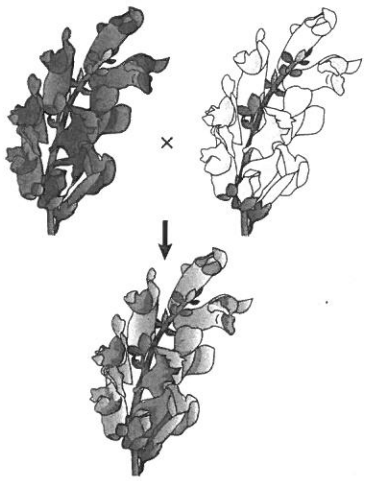


Codominance

Sometimes both alleles produce an effect in the phenotype; that is, neither allele is dominant or recessive to the other. The alleles are called codominant. These alleles are represented differently to the examples above. A capital letter is used for the gene, such as *C* for coat colour in cattle, and the alleles are shown using a superscript. For the cattle coat colour gene, the red coat allele is written as C^R and the white coat allele as C^W. A cow with the genotype C^RC^W will have a coat with both red and white hairs.

Incomplete dominance

If two different alleles produce a blended effect in the phenotype when they appear together, the mode of inheritance is called incomplete dominance. This is represented in a similar way to codominance. A capital letter is used for the gene, for example *F* for flower colour in plants, and then the alleles are shown using a superscript. The allele for red flowers is written as F^R and that for white flowers is F^W. A heterozygous individual would have a genotype of F^RF^W and a phenotype of pink flower colour (i.e. a blend, intermediate between red and white). Some resources may show this type of inheritance with only capital letter symbols.



There are many new definitions used when discussing genetics. To make sense of what you read and also to communicate scientifically, it is important to become familiar with the terms and learn their meanings.

Checkpoint 4

Compare the three types of inheritance.

Skill 3.3 review

1 Copy the table below into your notebook, then use the following information to **identify** the missing genotypes and phenotypes. The first example has been completed for you. The alleles used are:

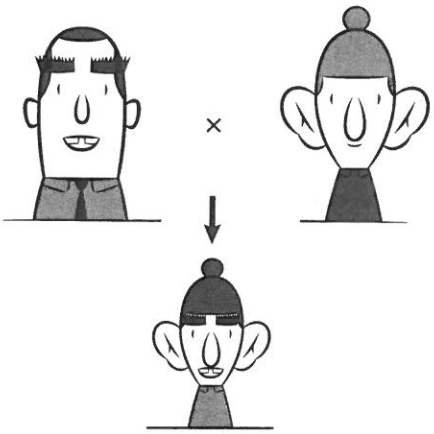
F = freckles *f* = no freckles *D* = dark hair *d* = light hair

B = brown eyes *b* = blue eyes *L* = long legs *l* = short legs

Genotype	Phenotype	Homozygous or heterozygous
DD *Bb*	Dark hair Brown eyes	Homozygous for dark hair Heterozygous for brown eyes
ll *Ff*		Homozygous for short legs Heterozygous for freckles
Ff *Bb*		
	Light hair Brown eyes	

3.4 Modelling simple inheritance

A Punnett square is a chart used by geneticists to model all of the possible genetic combinations from a cross between two parents whose alleles for a particular gene are known. It reveals the probability of offspring inheriting particular genetic characteristics. Punnett squares were named after English geneticist Reginald Punnett.



Using Punnett squares

Genetic problems can be clearly represented using a **Punnett square**. Th steps involved are outlined below, using pea colour as an example.

1 Identify the **gene** that you are investigating, and the possible **alleles** (variations of the gene). Assign letters to represent the alleles. Remember that capital letters represent **dominant** traits and lowercase letters represent **recessive** traits.

Alleles involved in inheritance of pea colour:

***Y* = allele for yellow peas**
***y* = allele for green peas**

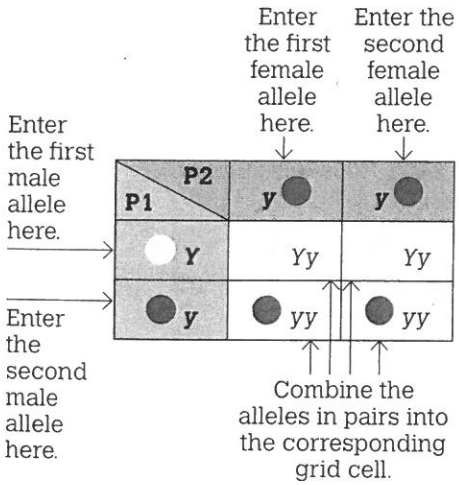

yellow


green

Genotype	Phenotype
YY	Yellow peas
Yy	Yellow peas
yy	Green peas

Parent 1: *Yy* **Heterozygous, yellow peas** × **Parent 2: *yy*** **Homozygous, green peas**

2 Write down the **genotypes** (alleles) and **phenotypes** (appearance) of each parent, as shown on the left. These will usually be given or possible to determine from the available information. Also state if the individual is **homozygous** (has two identical alleles) or **heterozygous** (has two different alleles).



3 Draw a Punnett square template like that shown on the left. To complete the Punnett square, follow the accompanying instructions. Note: when there is a dominant allele, it is always written first.

4 List the possible genotypes and phenotypes for offspring from the cross The allele combinations in each cell of the Punnett square represent the genotypes. These can then be used to determine the phenotypes. You may need to refer back to the alleles listed in step 1.

***Yy* = yellow peas** ***yy* = green peas**

5 Calculate the probability of each genotype and phenotype occurring i the offspring, as either a fraction or a percentage.

Of the four boxes in the Punnett square, two are *Yy* and two are *yy*. Therefore:

***Yy* = yellow peas 2/4 = 1/2 = 50%**

***yy* = green peas 2/4 = 1/2 = 50%**

In other words, in this cross there is a 50% chance of producing offspring with green peas and a 50% chance of offspring with yellow peas. This would be written below the Punnett square as:

Offspring: Probability is that 50% will be *Yy* (heterozygous for yellow peas) and 50% will be *yy* (homozygous for green peas).

Checkpoint 1

Refer to the pea colour cross on page 64 to answer the following questions.

a **Explain** which pea parent could be described as a **carrier**.

b **Propose** a parental cross that would result in all heterozygous yellow peas.

Skill 3.4 review

1 Wild coat colour in rabbits (*W*) is dominant to albino coat colour (*w*).

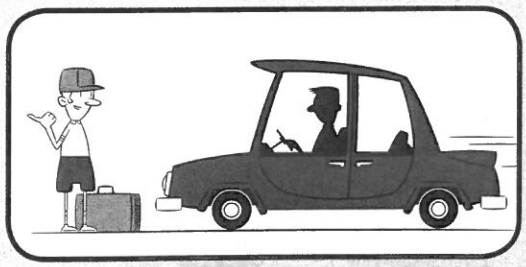
a A pure-breeding wild-coat rabbit is mated with an albino-coat rabbit. **Determine** the genotype and phenotype of the offspring.

b When two of the offspring are mated, **predict** what colour offspring you would expect and in what proportions. Use a Punnett square to set out your working.

c Suppose one of the offspring from part **a** was mated with an albino. **Predict** what colour offspring you would expect and in what proportions. Show your working.

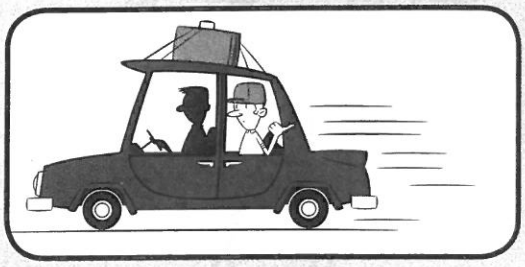
2 When considering thumb shape, straight thumb is dominant over hitchhiker's thumb. Draw a Punnett square for the cross below and **calculate** the probability of each possible genotype and phenotype in the resulting offspring.

Parent 1: Heterozygous for straight thumb × Parent 2: Heterozygous for straight thumb



3 When considering the ability to roll your tongue, tongue rolling is dominant over non-tongue rolling. Draw a Punnett square for the cross below and **calculate** the probability of each possible genotype and phenotype in the resulting offspring.

Parent 1: Homozygous for tongue rolling × Parent 2: Heterozygous for tongue rolling



4 A couple has calculated that the probability of them having a child with blue eyes is 50% and the probability of brown eyes is also 50%. They have already had two children, both with blue eyes. **Explain** the probability that their next child will also have blue eyes.